#15 · 2.8% of marks

Ch 13 — Business Valuation

SM TYK theory Qs: **EVA vs MVA · relative valuation is a "relative" value by comparative analysis — elaborate.** Numericals: NAV / NRV, P/E & capitalisation, DCF + TV, asset beta, relative (EBITDA multiple, EV), chop-shop, EVA, SVA, LBO, distressed firms, startup methods.

① Conceptual framework & terms (SM 1–2)

- → *First corporation valued / IPO: **Dutch East India Company** (United East India Company). Valuation = science + art; part of strategic FM*
- → ***Purposes (5)**: (a) information for internal stakeholders (b) comparison with similar enterprises — management efficiency (c) future **public listing** (d) strategic planning — value drivers, deploying surplus cash (e) **ball-park price** for acquisition*
- → ***PV** (₹1,000 in a year at 10% ≈ ₹909) · **IRR** (rate making NPV zero; higher IRR more likely selected) · **ROI**: (1,400 − 1,000) ÷ 1,000 = 40%; with additions use **average investment**: 1,000 + 200 during year, value 2,000 → 800 ÷ 1,100 = **72%**; historical ratio — course-correct only*
- → ***Gordon perpetual growth**: little used by investors — more parameters needed; dividends rarely grow steadily; "darling of academicians". **Terminal value** = "horizon value" at exit, usually by the Gordon model*

② Asset-based approach (SM 3.1)

*For M&A: unlisted → negotiated price; listed → market cap is a guide, but depends on market efficiency and may reflect only a small quoted portion. Three approaches: **asset, earnings, cash flow** (+ others). Asset approach values income-**generating assets**, not income; least useful for **IT companies** (IPR, people).*

Method	SM
***Net asset value** (book value)*	*Fixed assets + net current assets − long-term debt. Simple (historical cost) but balance sheet isn't a valuation device → **lower limit**; ignores intangibles (brand, IPR)*
***Net realisable value** (liquidation / adjusted book value)*	*Realisable value of assets − liquidation expenses − liabilities (expenses ignored if bought as going concern); buyer offers lowest prices; implies break-up; useful when acquirer will sell one part and integrate the rest. SM eg: NRV 1,05,000 ÷ 5,000 shares = **₹21***
Replacement value	*Cost of equivalent assets & liabilities in the open market; usually slightly above NRV; seen as the **maximum** an acquirer would pay; ignores staff loyalty*

*Conclusion: shows net worth fairly on going-concern basis but **ignores future revenue-generating ability** and market dynamics.*

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③ Income-based approach (SM 3.2)

For acquirers continuing the business; include extra earnings from the acquisition.

→ ***P/E (earning-yield multiplier)***: *price = EPS × P/E; for listed cos; captures expected growth & market expectations; for unlisted use peer / industry P/E. Gives the* ***minimum acceptable price*** *to target's holders. Steps: (i) P/E of equivalent quoted company (ii) adjust* ***down*** *for non-listing risk (iii) future* ***maintainable*** *EPS (iv) multiply*

→ ***Capitalisation of earnings***: *value = expected annual* ***maintainable profit*** *(weighted average of past profits, adjusted for synergy / scale) ÷ capitalisation rate (required earning yield = EPS ÷ price = 1 ÷ P/E). Forward-looking; but future profit hard to estimate, extraordinary items treated differently*

④ Cash-flow (DCF) approach (SM 3.3)

Free cash discounted to match risk; avoids over-reliance on history. ***5 steps***: *(a) free cash flows (b) forecast (c) discount rate from cost of capital (d) terminal value (e) PV of FCF + TV, interpret.*

SM eg: *EAT 600 − one-time income 200 + one-time expense 100 + depreciation 100 = FCF 600; grow 5%, less ΔWC & capex → 500 / 550 / 651.50; at 8% PV 463 / 471.35 / 517.29;* ***TV = 517.29 × 1.03 ÷ (0.08 − 0.03) = 10,656.17****; total* ***12,107.81****. Note: SM grows the year-3* ***PV****, so this TV figure is already a present value. TV is often the major part of value → validate g and WACC.*

⑤ Cost of equity (SM 4)

→ ***CAPM***: $R = r_f + \beta(r_m - r_f)$ — *popular but* ***over-simplifies*** *risk into two buckets*

→ ***APT*** *(Stephen Ross, 1970s): multiple macro factors, each with its own β:* $\mathbf{R_f + \beta_1 RP_1 + \beta_2 RP_2 + \ldots + \beta_n RP_n}$*; APT doesn't name the factors — analysts choose them*

⑥ Asset beta & unlisted firms (SM 4.3)

Private firms (often little debt) and a listed firm entering a ***new line of business*** *shouldn't use their own WACC → use* ***asset (ungeared) beta*** *= business risk only.*

$\beta_a = \beta_e \times E \div [E + D(1-t)] + \beta_d \times D(1-t) \div [E + D(1-t)]$*; with* $\beta_d = 0$: $\beta_a = \beta_e \times E \div [E + D(1-t)]$*; regear:* $\beta_e = \beta_a \times [E + D(1-t)] \div E$*.* $\beta_e \geq \beta_a$ *(equal if debt-free)*

Steps: *(1) find* ***pure-play*** *(proxy) firms and their* β_e *(2) de-gear each (3) one proxy → use it; several →* ***average*** *(or most appropriate) (4) re-gear at* ***own*** *capital structure (5) CAPM → required return / value*

Illus 1 *(X Pvt Ltd):* β_a *1.8, D:E 40:60, t 30% →* $\beta_e = 1.8[1 + 0.7 \times 40/60] =$ ***2.64***; $K_e = 5 + 2.64 \times 6 =$ ***20.84%***; *WACC = 20.84 × 0.6 + 12 × 0.7 × 0.4 =* ***15.864%***; *FCF 100/120/150 → value* ***272.15****. EBITDA 90 − extraordinary gain 10 − pending write-off 20 = 60 × 5 = EV* ***300****.*

⑦ Relative valuation (SM 5) — TYK Q2

- → Asset / earnings / cash-flow approaches give **intrinsic** value; relative valuation gives a **"relative" value by comparison with peers** — "valuation by multiples"; increasingly used to **validate** intrinsic value; aims at a value near market price (eg bond YTM close to the government index is more credible)
- → **4 steps**: (1) find the **drivers** for the multiple (2) compute via financial ratios (3) find **comparable firms** & compare (4) **iterate** to smooth deviations
- → **Step 1**: **EV multiples** — EV/EBITDA (popular), **EV/invested capital** (capital-intensive), **EV/sales** (cash-rich, big order book, organic growth) · **equity multiples** — P/E (celebrated), **PEG** = P/E ÷ growth (high-growth / sunrise industries)
- → **Step 2**: right ratio — exporter's value swings with FX (P/E misses Brexit-type shocks); EV/invested capital misfits asset-light firms or those with investment property
- → **Step 3** (hardest): no two firms alike — X (FTE pricing) vs Y (unit transfer pricing) differ in risk; niche firms → look across industries or **regression**; "take benchmarks with a pinch of salt"
- → **Assumptions (3)**: (i) market is **efficient** (ii) fundamentals–multiples link is **linear** (iii) comparables similar in structure, risk & growth. EV via entity value adjusted for debt, or balance-sheet approach

SM illustration	Answer
Illus 2: GP 10 L − indirect 4 L = EBITDA 6 L; 1 L shares; debt 3 L; surplus funds 5 L; r_f 4.5, r_m 12, β 0.9	(a) rate 4.5 + 0.9 × 7.5 = 11.25% → 600 ÷ 0.1125 = 5,333.33 thousand → **₹53.33** per share (b) 600 × 5 = 3,000 − 300 + 500 = equity **₹32 lakh**
Illus 3: H K Ltd 70,000 sh at ₹12; LT debt 2 L; cash 5 L	8.40 L + 2 L − 5 L = EV **₹5,40,000**

⑧ Contemporary approaches (SM 6.1)

- → Industry-specific metrics: **price per page visited** (internet), **price per subscriber** (online store) — still check cash-to-sales and DCF
- → **Goodwill-based**: asset value + goodwill as a multiple of sales or **footfall** (retail giant buying a mom-and-pop store)
- → **PER**: EPS 40, average price 50 → 1.25 (relative figure; use sector median)
- → **LBO by PE firms**: buy with **~70 : 30 debt : equity**, sell in 3–5 years. SM eg: EBITDA 100 → … Y3 EBDAT 112 × **exit multiple 7** = 784 − debt 100 = equity **684** (SM shows Y3 EBITDA 178; 20% growth on 144 would give 172.8). Risk: cyclical slowdown leaves heavy debt

9 Chop-shop / break-up (SM 6.2)

Find **undervalued multi-industry** firms worth more split up — buy assets below replacement value. **Steps**: (1) identify segments; average capitalisation ratios of pure-play firms (2) theoretical value per ratio (3) **average** the theoretical values. **Illus 4** (Cornett, market €4 m): cap/sales €4,005,000 · cap/assets €2,880,000 · cap/operating income €5,650,000 → average **€4,178,000** > €4 m.

10 Economic Value Added (SM 6.3)

- → Holistic: valuation and economic contribution; value is created only if returns **exceed the cost of capital**; separates operations from financing. Negative EVA = destroying value
- → **Why EVA**: others are number-crunching; EVA makes **management accountable** for decisions (eg rejecting a positive-IRR but slower-payback project shows up)

EVA = NOPAT − (Invested capital × WACC) = NOPAT − capital charge. NOPAT = EBIT − tax (depreciation stays as operating expense) + **non-cash** items (eg bad-debt provision). Invested capital = total assets − **non-interest-bearing current liabilities** (creditors, advance billing) + non-cash adjustment = equity + long-term debt, generally at **start** of year.

Illus 5: EBIT 410 − tax 123 + provision 20 = NOPAT **307**; IC = 1,300 − 400 + 20 = **920**; WACC = 800/900 × 8.45% + 100/900 × 12% × 0.7 = **8.44%**; charge 77.65 → EVA **229.35**.

11 Market Value Added (SM 6.4)

MVA = current market value of the firm − invested capital (eg 1,000 − 920 = 80). Addresses EVA issues (ignoring value drivers, book value); no basis for share value but gauges efficiency from **market-cap** view — value perceived by the market. EVA = **derived** value for discerning investors. High-MVA firms become market darlings / pricey; EVA and MVA may be briefly negatively correlated (eg ex-dividend) — gap closes.

TYK Q1	EVA	MVA
Measures	NOPAT − WACC × invested capital — internal, period performance	Market value − invested capital — market's view of future value creation
Based on	Accounting data adjusted (NOPAT, non-cash items)	Market capitalisation
Use	Management accountability; returns above hurdle rate	Perceived value added; not a basis for share value

12 Shareholder Value Analysis (SM 6.5)

NOPAT is historical and misses future investment opportunities → SVA uses **value drivers** (sales potential, investments, cost of incremental capital). **Steps**: (a) FCF from value drivers (b) discount at WACC (c) add TV (d) add market value of non-core assets & marketable investments (e) less debt = equity. SM: PVs 57.54 + 43.46 + 44.86 + TV (40.45 × 1/0.12 = 336.95) = 482.81 + investment property 35 − debt 19 = **498.81**.

13 Fair value (SM 7)

Buy at fair price — "no less no more". CA: arm's length; analyst: PV in cash; speculator: arbitrage. Booms defy fair value (dot-com). No single correct method → give a **range** (seller's minimum, buyer's maximum); deal by negotiation. **Exam approach**: (i) value by as many methods as data allows (ii) comment on each (iii) add any further information to the conclusion.

14 Going vs non-going concern (SM 8)

Going concern assumed (else different basis, disclosed). **Non-going concern = liquidation value** (assets sold, liabilities paid). Going-concern value = **total value** — usually higher: includes future profits, intangibles, goodwill (premium); liquidation also means lay-offs and reputation damage. Use non-going concern only when the firm has no value as a going concern.

15 Distressed companies (SM 9)

Distress = unable / struggling to pay creditors (high fixed costs, illiquid assets, cyclical revenue); can spread to operations; still may have value. **Why conventional methods fail**: TV assumes infinite life and growing CF (negative CF here) · negative, declining revenue — CF hard to estimate, DCF values a going concern, may give negative equity · discount rates assume sound firms → adjust for failure probability.

Method	SM
Modified DCF	Probability distribution of CF across outcomes, adjusted for default; K_e from updated D/E + unlevered β; K_d from updated default risk. Short-cut: **Expected CF_t = CF_t × (1 − probability of distress$_t$)**
DCF + distress value	(i) value as going concern on the path back to health (ii) probability of distress (iii) distress sale value (% of book or DCF value). **Equity = DCF equity × (1 − p) + distress sale value × p**
APV	**Firm value = unlevered value + (tax benefits of debt − expected bankruptcy cost)**; unlevered = FCFF at unlevered K_e; **expected bankruptcy cost = (unlevered value − distress sale value) × p**
Relative	Revenue & EBITDA multiples (P/E, P/BV often unusable); adjust down subjectively — telecom avg 2× revenue → **1.25×** for a distressed firm

16 Startups (SM 10)

Standard approaches (income, asset, market) presuppose an established, cash-generating business; startups have **no history**, value rests on untested growth and founders ("walking in the dark … night-vision goggles"). **Why traditional methods fail**: income — no / minimal positive CF · asset — mostly IP / intangibles, and going concern so not just realisable value · market — **disruptors** without established competitors (some market elements used in later rounds).

Value drivers: **product** (ready / prototype vs idea; market testing) · **management** (credentials, balance — over half of Indian unicorns have IIT / IIM founders) · **traction** · **revenue** streams · **industry attractiveness** (tourism amid lockdowns) · **demand-supply** of investors · **competitiveness** (first-mover; Ola after Uber).

Method	SM
Berkus (Dave Berkus)	5 success factors — basic value · technology · execution · strategic relationships · production & sales; caps **$2 m pre-revenue**, **$2.5 m post-revenue**; simple, ignores market factors
Cost-to-duplicate	All costs to build the startup / product incl. physical assets; criticised — ignores future revenue
Comparable transactions	"How much were similar startups valued at?" — XYZ ₹560 cr ÷ 24 cr users ≈ ₹23 / user × ABC's 1.75 cr ≈ **₹40 cr**; adjust for differentiators
Scorecard	Pre-revenue: average pre-money of peers × Σ(weight × score). Weights: team 0–30% · opportunity 0–25% · product 0–15% · competition 0–10% · marketing / channels 0–10% · need for more investment 0–5% · others 0–5%. Eg marketing 10% × 150% = 0.15
First Chicago	DCF + market: worst / normal / best case values × probabilities → weighted value
Venture capital	Investor's target multiple (10×, 20×, 30×) or IRR; discount exit value to get **post-money** value

17 Digital platforms (SM 11)

- → Software infrastructure for many-to-many interactions: **marketplace** (Booking.com, Uber, Amazon) · **search** (Google, Bing, Baidu) · **repository** (Spotify, YouTube, GitHub) · **communication** (WhatsApp, Teams, Telegram, Slack) · **community** (Facebook, LinkedIn) · **payments** (Paytm, GPay)
- → **Income**: **top-down** — TAM → SAM → SOM, time to reach share (ambitious for nascent firms) · **bottom-up** — from limited resources. Cash burn early. FCFF at WACC, FCFE at K_e; beta hard (few comparables) → add **company-specific risk premium (CSRP / alpha)**, higher for nascent platforms
- → **Market**: comparables scarce, profits / EBITDA negative, capital-light (low BV) → compare on **revenue drivers** (bookings, users, merchants, views per user, search speed) · **Cost**: hours of coding — ignores revenue capacity

Professional / consultancy firms (SM 12): compare historical data with industry KPIs and competitors (audited statements, tax returns); factor risk in projected growth (TV); **normalise** income and cash flows (add back non-cash and firm-specific items).

18 Impact of ESG on valuation (SM 13)

- → **E**: climate change, water, waste, emissions, biodiversity · **S**: employee development, diversity & inclusion, community, health & safety, customer · **G**: board independence & diversity, anti-bribery, tax transparency, ethical conduct
- → Developments: ESG funds > $50 bn in 2020; ESG assets > $35 trillion · green bonds crossed **$1 trillion** (2020) · sustainability taxonomies (EU) · IFRS single global ESG standard · **SEBI** proposed ESG disclosure framework (Feb 2023)
- → High ESG → **cheaper debt**, access to green / social / sustainability-linked bonds; ESG moved from "good to have" to "must have"
- → **In valuation**: adjust the discount rate (practical, less explicit) or **cash flows** (more explicit): **E** — 2-degree scenario analysis, carbon points · **S** — cost of labour conditions, CSR, welfare · **G** — penalties, fines, taxes from poor governance

19 Case studies

Vodafone–Idea: market value, comparable multiples, NAV (31 Dec 2016); **no DCF** (no projections provided); share-exchange basis; Idea fell up to 14.57% — "**perceived value**" isn't quantifiable. **Facebook–WhatsApp**: $21.8 bn; 450 m users (70% active), +1 m a day; ~**$55 per user** incl. $4 bn retention payout.

20 Correlation — how Ch 13 actually connects

Ch 13 idea	The actual link (app-written cross-reference; the Ch 13 side is SM)
Asset beta, re-gearing	**Ch 11**: SM Nepal illustration uses the competitor's beta the same way; **Ch 3**: APV needs the unlevered K_e from here.
DCF, FCFF / FCFE, EV	**Ch 5**: the FCFF / FCFE formulas and EV types live there; TV = Gordon model.
Valuation range	**Ch 14**: seller's minimum and buyer's maximum set the swap-ratio range; chop-shop is the M&A "break-up" logic.
APT, CAPM	**Ch 6**: APT surprise formula, CAPM / SML — the K_e inputs.
EVA	**FR Ind AS 103 / 36**: value in use and goodwill impairment use the same discounted-cash-flow idea; EVA is the management-performance view.
Startup & VC methods	**Ch 15**: VC buy-back IRR of 18–25% = the VC method's target rate.
Distressed APV	**Ch 3 / Ch 11**: same APV, with expected bankruptcy cost subtracted.